MSDI Calculator

Operational Cost

Methodology

Technical Methodology Memorandum

Prepared for internal model documentation and technical review

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Model scope: Manure Sub-Surface Drip Irrigation (mSDI-e) economic screening calculator

Document status: MSDI calculator operational cost logic and methodology.

Purpose and Scope

This memorandum documents the operational expense methodology used by the MSDI Calculator to compare annual operating costs for an existing dairy manure and irrigation system against a proposed manure subsurface drip irrigation (mSDI-e) scenario. The methodology translates user inputs, lookup-table coefficients, and upstream model outputs into annual cost line items. It is intended for technical review, model documentation, and future maintenance of the application codebase.

Relationship to Other Methodologies

This operational-cost methodology depends on several previously documented model components. Weather and crop-water methods establish seasonal irrigation demand. Manure coefficients and manure-application equipment performance methods define manure-stream transformations and application rates. Capital-cost methodology documents installed mSDI-e costs separately. This memorandum is limited to recurring annual operating expenses and does not restate those upstream methodologies except where their outputs enter OPEX calculations.

Supporting methodology	Role in OPEX calculations
Weather Data Methodology	Supplies ET0 and precipitation values used by crop water demand and mSDI-e water-use estimates.
Crop Water Demand and Irrigation Requirement Methodology	Supplies seasonal mSDI-e runtime and water-only applied volumes used in water, labor, utility, and fuel calculations.
Manure Management Coefficients Methodology	Supplies manure mass-balance and nutrient-concentration logic used by manure and synthetic fertilizer calculations.
Manure Application Equipment Performance Methodology	Supplies application rate, fuel rate, and electric-use coefficients for manure application OPEX.
Capital Cost Methodology	Documents capital expenditure and simple annualized capital cost; capital costs are displayed alongside OPEX but are not recurring OPEX line items.

Terminology and Units

Term	Meaning in this methodology	Primary units
Existing system	User-reported current irrigation, manure application, tillage, weed, and fertilizer system applied to the acres considered for mSDI-e conversion.	Annual quantity/cost
mSDI-e scenario	Modeled proposed manure subsurface drip irrigation scenario, including modeled mSDI water use, manure application through the system, and residual traditional field operations.	Annual quantity/cost
Delta cost	mSDI-e cost minus existing-system cost. Positive values indicate increased expense; negative values indicate savings.	USD/year
OPEX item	A row-level calculation returning existing quantity, mSDI-e quantity, existing cost, mSDI-e cost, and delta cost.	varies
Acres for SDI	Acreage selected by the user for prospective mSDI-e installation and used to scale annual OPEX line items.	acres
Synthetic N	Commercial nitrogen fertilizer required after accounting for manure-derived plant-available nitrogen.	lb N/year
Manure application quantity	Traditional non-SDI manure application still required in each scenario, expressed as wet tons.	wet tons/year

Operational Cost Framework

The OPEX calculation engine represents each operational expense as a row-level object with existing and mSDI-e quantities and costs. The shared object builder returns the row identifier, label, units, existing quantity, mSDI-e quantity, quantity difference, existing cost, mSDI-e cost, and cost difference. The calculation pattern is consistent across all OPEX rows:

Equation 1. Delta Quantity = mSDI-e Quantity - Existing Quantity

Equation 2. Delta Cost = mSDI-e Cost - Existing Cost

Equation 3. Total Existing OPEX = sum(Existing Cost_i) for all OPEX rows

Equation 4. Total mSDI-e OPEX = sum(mSDI-e Cost_i) for all OPEX rows

Equation 5. Annual Net Expense Change = Total mSDI-e OPEX - Total Existing OPEX

Equation 6. Per-Acre Cost = Annual Cost / Acres for SDI

The implementation constructs all row-level OPEX items and then aggregates them into total existing cost, total mSDI-e cost, total delta cost, and per-acre values. Report display components then group several row-level deltas into user-facing summary categories, such as irrigation pumping costs, weed management costs, manure and fertilizer application costs, and tillage costs.

Report category	Row-level components included
Irrigation Water Assessments	Irrigation water row only.
Irrigation Pumping Costs	Irrigation utility delta plus irrigation fuel delta.
Irrigation Labor Costs	Irrigation labor row only.
Weed Management Costs	Herbicide material, herbicide fuel, and herbicide labor deltas.
Commercial Fertilizer Costs	Synthetic fertilizer quantity row cost delta.
Manure and Fertilizer Application Costs	Manure application fuel, utility, and labor deltas.
Tillage Costs	Tillage fuel and tillage labor deltas.
Annual Net Expenses	Sum of all row-level OPEX cost deltas plus annualized capital cost when displayed in the final results table.

Irrigation Water Cost Methodology

The irrigation water OPEX row estimates acre-feet of water used under the existing system and under mSDI-e, then applies user-provided water prices and fixed assessments. Existing water use is derived from the reported average annual primary and secondary water use per acre and scaled to the mSDI-e acreage.

Equation 7. Existing Water Use (AF/year) = ((Primary Annual Water Use + Secondary Annual Water Use) / 12) x Acres for SDI

The mSDI-e water quantity is calculated from upstream background calculations for seasonal water-only volumes. Seasonal gallons are summed and converted to acre-feet using the gallons-per-acre-foot coefficient.

Equation 8. mSDI-e Water Use (AF/year) = (Spring Water Gallons + Summer Water Gallons + Fall Water Gallons + Winter Water Gallons) / Gallons per Acre-Foot

Water cost is calculated by weighting the total water quantity between primary and secondary water sources according to their reported shares of total water use. Variable water prices and fixed per-acre assessments are applied to each source.

Equation 9. Primary Share = Primary Annual Water Use / (Primary Annual Water Use + Secondary Annual Water Use)

Equation 10. Secondary Share = Secondary Annual Water Use / (Primary Annual Water Use + Secondary Annual Water Use)

Equation 11. Water Cost = (Water Quantity x Primary Share x Primary Price + Acres x Primary Fixed Assessment) + (Water Quantity x Secondary Share x Secondary Price + Acres x Secondary Fixed Assessment)

If no water use is reported for either source, water cost is set to zero to avoid division by zero. Groundwater and surface-water unit prices and fixed assessments are selected based on the user-designated water source type.

Part 1. Irrigation Pumping Fuel, Electricity, and Labor

1.1 Irrigation Fuel

Existing irrigation fuel use is based on the user-reported fuel use per acre and scaled to the mSDI-e acreage. The mSDI-e fuel row uses the SDI fuel-gallons-per-hour coefficient. In the current implementation this coefficient can be zero where the modeled mSDI-e pumping system is electric-powered.

Equation 12. Existing Irrigation Fuel (gal/year) = Existing Fuel Use per Acre x Acres for SDI

Equation 13. Existing Irrigation Fuel Cost = Existing Irrigation Fuel x Fuel Price

Equation 14. mSDI-e Irrigation Fuel (gal/year) = SDI Fuel gal/hr x Acres for SDI

Equation 15. mSDI-e Irrigation Fuel Cost = mSDI-e Irrigation Fuel x Fuel Price

1.2 Irrigation Electricity

Existing electricity use is based on user-reported electricity use and scaled to the mSDI-e acreage. mSDI-e electricity is calculated from seasonal runtime, number of sets, weeks per season, and the SDI pump kWh coefficient.

Equation 16. Existing Irrigation Electricity (kWh/year) = Existing Electricity Use x Acres for SDI

Equation 17. Existing Irrigation Electricity Cost = Existing Irrigation Electricity x Electricity Price

Equation 18. Weeks per Season = Days in Season / Days in Week

Equation 19. mSDI-e Electricity = sum(max(Season Runtime, 0) x Weeks per Season x Number of Sets x SDI Pump kWh)

Equation 20. mSDI-e Electricity Cost = mSDI-e Electricity x Electricity Price

1.3 Irrigation Labor

Existing irrigation labor is allocated from total annual labor hours over total acres and scaled to acres for SDI. mSDI-e irrigation labor is calculated from modeled seasonal runtimes, a labor-per-run-hour coefficient, number of sets, and rodent-maintenance hours where applicable.

Equation 21. Existing Irrigation Labor (hr/year) = (Annual Irrigation Labor Hours / Total Acres) x Acres for SDI

Equation 22. Base mSDI-e Labor = sum(max(Season Runtime, 0) x Weeks per Season x Labor per Run Hour x Number of Sets)

Equation 23. Rodent Maintenance Labor = Rodent Maintenance hr/ac/year x Acres for SDI

Equation 24. mSDI-e Irrigation Labor = Base mSDI-e Labor + Rodent Maintenance Labor

Equation 25. Irrigation Labor Cost = Irrigation Labor Hours x Hourly Labor Cost

Part 2. Manure Quantity and Commercial Nitrogen Methodology

2.1 Existing Traditional Manure Application Quantity

The existing manure application quantity is calculated from expected weekly manure liquid volume, manure density, the user-reported percentage of liquid manure applied, the share of total acres represented by acres for SDI, and weeks per year.

Equation 26. Tons per Week = Expected Manure Liquid gal/week x Density lb/gal / Pounds per Ton

Equation 27. Existing Traditional Manure Tons = Tons per Week x (% Liquid Manure Applied / 100) x (Acres for SDI / Total Acres) x Weeks per Year

2.2 mSDI-e Residual Traditional Manure Quantity

Under the mSDI-e scenario, traditional non-SDI manure application is retained only where seasonal crop nitrogen demand remains after accounting for manure delivered through the mSDI-e system. The amount of remaining traditional manure is capped by the amount of traditional manure nitrogen that would otherwise be available in that season.

Equation 28. Seasonal Crop N Demand = Seasonal N Demand lb/ac x Acres for SDI

Equation 29. mSDI-e Manure N Supplied = mSDI-e Seasonal Manure Gallons x Density lb/gal x Seasonal TN Fraction x Plant-Available TN Fraction

Equation 30. Remaining N Demand after mSDI-e = max(0, Seasonal Crop N Demand - mSDI-e Manure N Supplied)

Equation 31. Existing Traditional N Available = Seasonal Weekly Manure Volume at Terminus x Weeks per Season x (% Liquid Manure Applied / 100) x Acres Ratio x Density x Seasonal TN Fraction x Plant-Available TN Fraction

Equation 32. Traditional N Used = min(max(0, Remaining N Demand), max(0, Existing Traditional N Available))

Equation 33. Traditional Manure Tons Needed = (Traditional N Used / (Density x Seasonal TN Fraction x Plant-Available TN Fraction)) x Density / Pounds per Ton

Equation 34. mSDI-e Traditional Manure Tons = sum(Seasonal Traditional Manure Tons Needed)

2.3 Commercial Synthetic Nitrogen

Existing commercial nitrogen is calculated as the greater of a model-calculated N deficiency and a user-provided minimum or override value. The model-calculated deficiency equals annual crop N demand on the SDI acres less plant-available nitrogen supplied by existing traditional manure application.

Equation 35. Total Crop N Demand = Spring N Demand + Summer N Demand + Fall N Demand + Winter N Demand

Equation 36. Existing Manure N = Existing Manure Tons x Pounds per Ton x Average Seasonal TN Fraction x Plant-Available TN Fraction

Equation 37. Calculated Existing Synthetic N = Total Crop N Demand - Existing Manure N

Equation 38. User Synthetic N = User-Reported Synthetic N lb/ac x Acres for SDI

Equation 39. Existing Synthetic N = max(Calculated Existing Synthetic N, User Synthetic N)

Equation 40. Existing Synthetic Fertilizer Cost = Existing Synthetic N x Synthetic N Cost per lb

For the mSDI-e scenario, the calculation is performed seasonally. Each season subtracts manure N delivered by mSDI-e and N available through remaining traditional manure application. Any unmet seasonal N demand is assigned to commercial synthetic nitrogen, with negative values prevented.

Equation 41. Seasonal mSDI-e Synthetic N = max(0, Remaining N Demand after mSDI-e - Traditional N Used)

Equation 42. mSDI-e Synthetic N = sum(Seasonal mSDI-e Synthetic N)

Equation 43. mSDI-e Synthetic Fertilizer Cost = mSDI-e Synthetic N x Synthetic N Cost per lb

Part 3. Manure and Fertilizer Application Cost Methodology

Manure application fuel, labor, and utility calculations use a shared application workload. Application rate is retrieved from the manure application lookup table for the user-selected manure application method. Existing workload includes traditional manure tons plus synthetic fertilizer tons; mSDI-e workload includes only the residual traditional manure tons that remain after mSDI-e manure delivery. This treatment prevents synthetic fertilizer from being counted as traditional spreading workload in the mSDI-e scenario.

Equation 44. Application Rate = Manure_App_LU app_rate (tons/hour)

Equation 45. Existing Application Tons = Existing Manure Tons + Existing Synthetic N lb / Pounds per Ton

Equation 46. mSDI-e Application Tons = mSDI-e Traditional Manure Tons

Equation 47. Application Hours = Application Tons / Application Rate, where Application Rate > 0

3.1 Application Fuel

Equation 48. Existing Manure Application Fuel = Existing Application Hours x Manure Application Fuel Rate

Equation 49. Existing Manure Application Fuel Cost = Existing Manure Application Fuel x Fuel Price

Equation 50. mSDI-e Manure Application Fuel = mSDI-e Application Hours x SDI Fuel gal/hr

Equation 51. mSDI-e Manure Application Fuel Cost = mSDI-e Manure Application Fuel x Fuel Price

3.2 Application Electricity

The manure application utility row uses manure-only application workload on both sides. It calculates application hours from manure tons and the manure application rate, then multiplies by the manure-application electric-use coefficient.

Equation 52. Manure-Only Application Hours = Manure Tons / Application Rate

Equation 53. Manure Application Electricity = Manure-Only Application Hours x Manure Application Electric Rate

Equation 54. Manure Application Electricity Cost = Manure Application Electricity x Electricity Price

3.3 Application Labor

Equation 55. Existing Manure Application Labor Cost = Existing Application Hours x Hourly Labor Cost

Equation 56. mSDI-e Labor Factor = 1 if SDI Fuel gal/hr > 0; otherwise 0

Equation 57. mSDI-e Manure Application Labor Cost = mSDI-e Application Hours x Hourly Labor Cost x mSDI-e Labor Factor

Tillage Cost Methodology

Tillage labor is calculated using tillage hours per acre per crop from the tillage lookup table. Existing tillage uses the user-selected tillage strategy. The mSDI-e scenario assumes conservation tillage. Tillage fuel is calculated from tillage labor hours and the tractor fuel-use coefficient.

Equation 58. Existing Tillage Labor = Existing Tillage hr/ac x Acres for SDI

Equation 59. mSDI-e Tillage Labor = Conservation Tillage hr/ac x Acres for SDI

Equation 60. Tillage Labor Cost = Tillage Labor Hours x Hourly Labor Cost

Equation 61. Tillage Fuel = Tillage Labor Hours x Gallons Fuel per Tractor Hour

Equation 62. Tillage Fuel Cost = Tillage Fuel x Fuel Price

Weed Management and Herbicide Cost Methodology

Herbicide material and application labor are calculated from the weed pressure lookup table. Existing values use the user-selected weed-pressure category. The mSDI-e scenario assumes low weed pressure because subsurface application reduces surface wetting and may reduce weed germination pressure. Herbicide fuel is calculated from herbicide labor hours and the tractor fuel-use coefficient.

Equation 63. Existing Herbicide Material Cost = Existing Weed Material Cost per Crop x Acres for SDI

Equation 64. mSDI-e Herbicide Material Cost = Low-Pressure Weed Material Cost per Crop x Acres for SDI

Equation 65. Existing Herbicide Labor = Existing Weed Management hr/ac/crop x Acres for SDI

Equation 66. mSDI-e Herbicide Labor = Low-Pressure Weed Management hr/ac/crop x Acres for SDI

Equation 67. Herbicide Labor Cost = Herbicide Labor Hours x Hourly Labor Cost

Equation 68. Herbicide Fuel = Herbicide Labor Hours x Gallons Fuel per Tractor Hour

Equation 69. Herbicide Fuel Cost = Herbicide Fuel x Fuel Price

Lookup Tables and User Inputs

Input or lookup source	Use in OPEX methodology
User form: acres_for_sdi	Scales most annual quantities and costs to the acreage being evaluated for mSDI-e.
User form: total_acres	Allocates whole-farm annual quantities to the SDI acreage where needed.
User form: water use, water source, water price, fixed assessments	Used to calculate existing and mSDI-e irrigation water costs.
User form: fuel cost, electricity cost, hourly labor cost	Unit prices for fuel-, electricity-, and labor-based OPEX rows.
lookup_coefficients	Conversion constants, weeks per year, days per season/week, density, plant-available TN fraction, fuel and pump-use coefficients.
lookup_manure_app	Manure application rate, fuel rate, and electric-use rate.
lookup_tillage	Hours per acre per crop for existing tillage strategy and conservation tillage.
lookup_weeds	Herbicide material cost and application labor hours under weed-pressure categories.
lookup_rodents	Rodent maintenance hours per acre per year for mSDI-e irrigation labor.
Background calculations	Seasonal runtime, seasonal manure volumes, seasonal TN percentages, and seasonal crop N demand.

Assumptions and Limitations

- The methodology is intended for screening-level economic analysis, not site-specific engineering design or audited financial planning.
- User-reported prices and baseline quantities are assumed to be representative of current annual operating conditions.
- Existing-system costs are allocated to acres for SDI using proportional scaling where the user reports whole-farm values.
- mSDI-e irrigation fuel may be zero where the modeled system is assumed electric-powered; electricity is then accounted for through SDI pump kWh.
- mSDI-e tillage is assumed to use conservation tillage, while existing tillage follows the user-selected current strategy.
- mSDI-e weed pressure is assumed to be low pressure for herbicide material and labor calculations.
- Manure application costs depend strongly on equipment, haul distance, field layout, solids content, and operator efficiency; the lookup rates represent screening-level assumptions.
- Synthetic fertilizer calculations focus on nitrogen and do not independently price phosphorus, potassium, or micronutrient amendments.
- The manure application utility row uses manure-only workload; fuel and labor rows use the shared application workload defined in the implementation.
- All costs are annual operating expenses unless explicitly identified as capital costs in a separate methodology.

Processing Workflow Summary

1. Collect user form inputs for acreage, water use, water source, prices, labor, fuel, electricity, manure application, tillage, weed pressure, and manure/nutrient management.
2. Retrieve constants and lookup coefficients from generated JSON lookup tables.
3. Build individual OPEX rows for irrigation water, fuel, utility use, labor, manure quantity, synthetic N, manure application fuel/utility/labor, tillage fuel/labor, and herbicide material/fuel/labor.
4. For each row, calculate existing quantity and cost, mSDI-e quantity and cost, and delta cost.
5. Aggregate all row-level costs into total existing OPEX, total mSDI-e OPEX, total delta OPEX, and per-acre values.
6. Group row-level deltas into report-facing categories for the results page, PDF export, and CSV export.

Implementation Traceability

Methodology section	Primary implementation files
Irrigation water costs	row-03-irrigation-water.ts; water-utils.ts
Irrigation fuel	row-04-irrigation-fuel.ts
Irrigation labor	row-05-irrigation-labor.ts
Irrigation utility/electricity	row-06-irrigation-utility.ts
Manure quantity	row-07-manure-quantity.ts
Synthetic nitrogen	row-08-synthetic-fertilizer.ts
Manure application fuel/utility/labor	row-09-manure-fuel.ts; row-10-manure-utility.ts; row-11-manure-labor.ts; manure-application-utils.ts
Tillage	row-12-tillage-fuel.ts; row-13-tillage-labor.ts
Herbicide/weed management	row-14-herbicide-material.ts; row-15-herbicide-fuel.ts; row-16-herbicide-labor.ts
Aggregation and report grouping	result-utils.ts
Coefficient and lookup access	coefficients.ts; lookup-utils.ts

References and Source Materials

This memorandum is based primarily on the MSDI Calculator implementation files and previously prepared coefficient methodologies. Source files reviewed for this document include the OPEX row calculation files, result aggregation utilities, lookup utilities, and coefficient accessors listed in Section 14. External literature supporting individual coefficient values is documented in the corresponding coefficient-specific memoranda, including the Manure Application Equipment Performance Methodology, Crop and Soil Coefficients Methodology, and Manure Management Coefficients Methodology.

Appendix A. OPEX Row Summary

OPEX row	Quantity basis	Cost basis
Irrigation Water	Existing AF from reported annual water use; mSDI-e AF from modeled seasonal water-only gallons.	Weighted primary/secondary water price plus fixed assessments.
Irrigation Fuel	Existing gallons from fuel use per acre; mSDI-e gallons from SDI fuel coefficient.	Gallons x fuel price.
Irrigation Labor	Existing annual labor allocated to SDI acres; mSDI-e runtime labor plus rodent maintenance.	Labor hours x hourly labor cost.
Irrigation Utility	Existing kWh from user input; mSDI-e kWh from runtime, sets, and pump kWh coefficient.	kWh x electricity price.
Manure Quantity	Existing traditional manure tons; residual traditional manure tons after mSDI-e manure N contribution.	Quantity row only; no direct cost assigned.
Synthetic Fertilizer	Existing and mSDI-e synthetic N requirement after manure N contributions.	lb N x synthetic N cost per lb.
Manure Fuel	Application hours x fuel rate for existing; SDI fuel coefficient for mSDI-e residual workload.	Gallons x fuel price.
Manure Utility	Manure-only application hours x electric-use rate.	kWh x electricity price.
Manure Labor	Application hours for existing and residual mSDI-e traditional manure workload.	Hours x hourly labor cost, adjusted by SDI labor factor.
Tillage Fuel	Tillage labor hours x tractor fuel coefficient.	Gallons x fuel price.
Tillage Labor	Existing strategy hours/ac; mSDI-e conservation tillage hours/ac.	Hours x hourly labor cost.
Herbicide Material	Existing weed-pressure material cost/ac; mSDI-e low-pressure material cost/ac.	Material cost directly.
Herbicide Fuel	Herbicide labor hours x tractor fuel coefficient.	Gallons x fuel price.
Herbicide Labor	Existing weed-pressure hours/ac; mSDI-e low-pressure hours/ac.	Hours x hourly labor cost.